Biomimetic fish robot development roadmap

"Identify open research problems and unexplored intersections between these technologies.. Compare existing systems and distinguish experimentally validated results from simulations and proposed concepts. Propose a phased development roadmap from simulation to prototype, with suitable performance metrics, baseline comparisons, validation experiments, and relevant papers with DOIs."
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The main open problem is not any single component, but the lack of an end-to-end validated stack that combines fish-inspired propulsion, low-noise/disturbance validation, physics-informed fluid modeling, and modern autonomy/control in one robot fish platform. Current work is strongest when it validates one layer well—propulsion or control or sensing—but the intersections among **simulation tools, soft actuation, ROS 2-style autonomy, and ecological disturbance metrics** remain largely unexplored or only proposed [127] [131] [139] [140] [141] [142].

1) What is experimentally validated vs. what is still simulation/proposal?

Category	Representative systems	What is validated	What remains weak
Experimentally validated prototypes	Multi-objective CPG robotic fish [1], undulating fin robot [143], bionic ocean sunfish dual-fin robot [144], soft fish with fluidic actuation [122], smart biomimetic fish robot with navigation and obstacle avoidance [64]	Swimming performance, trajectory/control behavior, or specific mechanical functions were tested on hardware [1] [64] [122] [143] [144]	Most experiments are in controlled pools/tanks, with limited long-duration field validation and limited ecological disturbance measurements [64] [139] [140]
Simulation-supported systems with partial experiments	Geometry optimization and hydrodynamic models [145] [146] [147] [148], 3DGS+CFD modeling [149], physics-informed DRL in vortex streets [150]	Models are often cross-checked against experiments and trends usually match [145] [146] [147] [148]	Simulations still idealize friction, surface roughness, structural deformation, and turbulence; simulation-to-reality gaps remain explicit [139] [145] [148] [150]

Category	Representative systems	What is validated	What remains weak
Proposed or still-maturing concepts	ROS 2-style integrated autonomy, PINN-driven CFD design loops, fully integrated soft robotic fish monitoring stacks	Conceptual fit is strong, and related components exist separately [140] [141] [142]	No retrieved study demonstrates a single integrated platform that jointly delivers soft actuation, ROS 2-native autonomy, PINN-based hydrodynamic optimization, and ecological disturbance validation [140] [141] [142]

2) Open research problems and unexplored intersections

A. Low-disturbance monitoring still lacks direct proof

Biomimetic fish robots are repeatedly described as quiet, flexible, and low-disturbance alternatives to propeller vehicles, but the evidence base is still mostly indirect or performance-oriented rather than ecology-oriented [131] [142] [147] [151]. The biggest gap is **direct measurement of acoustic output, wake structure, and biological response in realistic deployments** [139] [140].

B. Simulation is ahead of validation

Many studies validate CFD or dynamic models against experiments, but the agreement often depends on simplifying assumptions such as fixed Reynolds dependence, perfectly smooth surfaces, or neglected deformation/friction [1] [145] [148]. That means a major research problem is **closing the sim-to-real gap for coupled fluid–structure–control behavior** [139] [140] [150].

C. Soft robotics is promising but not yet deployment-ready

Soft fish concepts can improve biomimicry and adaptivity, and variable-stiffness mechanisms are repeatedly shown to matter for performance [122] [152] [153]. However, the literature still points to unresolved issues in **proprioception, waterproofing, fatigue, pressure sensitivity, and controller co-design** [127] [140] [154].

D. Control and sensing are under-integrated

There is strong evidence for CPG-based, MPC-based, RL-based, and adaptive control, but most systems still address control as a separate layer from hydrodynamics and sensing [1] [150] [155] [156] [157]. Proprioceptive and lateral-line-inspired sensing are identified as important, yet they are rarely fused into a unified autonomy stack [127] [147].

E. ROS 2 and physics-informed modeling are an unexplored combination

ROS 2 is explicitly identified as a largely unresolved middleware-level challenge for physics-based monitoring in heterogeneous robotic systems, while PINN-based methods are still struggling with turbulence, scaling, and training stability [141]. The unexplored intersection is **ROS 2 + online hydrodynamic surrogates/PINNs for real-time fish-robot adaptation** [141].

3) Phased development roadmap: simulation to prototype

Phase	Goal	Core metrics	Baselines	Validation experiments
Phase 1: Digital model	Build a coupled propulsion–control–fluid model	Speed, thrust, efficiency, Strouhal number, power cost, trajectory error, wake metrics [1] [146] [150]	Quasi-steady dynamic model, baseline CPG, PID, standard CFD [1] [158]	Compare simulated speed/head angle vs. model predictions; evaluate sensitivity to Reynolds number, stiffness, fin geometry, and actuation frequency [1] [145] [146]
Phase 2: Hydrodynamics and noise	Optimize shape and fin kinematics for low disturbance	BL/s, thrust coefficient, efficiency, wake coherence, noise/SPL, turbulence intensity [147] [152] [159] [160]	Rigid fin, motorized tail, propeller-based UUV [142] [147] [152]	CFD/PIV/hydroacoustic tests on fin geometry and stiffness variants; compare with baseline robotic fish and propeller vehicles [145] [147] [159]
Phase 3: Control integration	Add closed-loop autonomy and disturbance rejection	Tracking error, response time, robustness, energy per distance, recovery from perturbations [150] [155] [156] [157]	PID, CPG-only, DDPG, PPO, DWA, backstepping [150] [156] [158]	Benchmark on identical trajectories and disturbance scenarios using fair comparison protocols [157] [158]
Phase 4: Prototype with sensing	Add proprioception and environmental sensing	Navigation accuracy, obstacle avoidance success, leakage/stability, sensing latency, data quality [64] [127] [153] [161]	Sensorless or teleoperated prototypes [64] [131]	Tank tests with obstacle courses, real-time telemetry, flow sensing, and repeated starts/stops [64] [153]
Phase 5: Field validation	Demonstrate low-disturbance monitoring in realistic water	Mission duration, energy autonomy, target detection, ecological response, acoustic footprint [140] [162]	Tank/pool results only [64] [122]	Long-duration trials near fish or sensitive habitat, with fish-behavior, wake, and acoustic measurements [140] [163]

4) Best candidate baselines for comparison

- **Propulsion baselines:** rigid-link robotic fish, motor-driven tails, propeller UUVs, and smart-material fish with lower torque but quieter operation [142] [152] [164].
- **Control baselines:** PID, classic CPG, backstepping, SMC, DDPG, PPO, DWA, and NMPC [150] [155] [156] [157] [158] [165].
- **Simulation baselines:** quasi-steady fluid models, standard CFD, and non-deformable fin assumptions [1] [145] [148].
- **Soft-robotics baselines:** rigid fish robots, simple compliant fins, and single-actuator soft platforms [122] [152] [153].

5) Papers to anchor the roadmap

- **Validated control/prototype:** *Multi-objective evolutionary design of central pattern generator network for biomimetic robotic fish* — DOI **10.1007/s40747-022-00883-7** [1]
- **Simulation-to-experiment propulsion optimization:** *Optimization of swimming mode for elongated undulating fin using multi-agent deep deterministic policy gradient* — DOI **10.1016/j.jestch.2024.101783** [143]
- **Experimental soft actuation:** *Modeling and Dynamic Analysis of Fish Robot with Soft Fluidic Actuation* — DOI **10.1007/s10846-024-02196-y** [122]
- **Validated sensing + control prototype:** *Development of a Smart Biomimetic Fish Robot Integrating Autonomous Navigation, Obstacle Avoidance, and Real-Time Motion Control* — DOI **10.1109/ICSMech69390.2026.11647206** [64]
- **Validated low-noise / disturbance-oriented designs:** *Hydrodynamic pressure sensing for a biomimetic robotic fish caudal fin integrated with a resistive pressure sensor* — DOI **10.1088/1748-3190/ad6d21** [147]
- **Roadmap-level synthesis:** *From Biological Mechanisms to Task-Oriented Design Principles: A Critical Review of Fish-like Biomimetic Robots* — DOI **10.3390/biomimetics11090630** [140]
- **Simulation/control integration:** *High-efficiency autonomous swimming control of robotic fish in vortex streets: A physics-informed DRL framework* — DOI **10.1016/j.compfluid.2026.107056** [150]
- **ROS 2 intersection candidate:** *Physics-Based Attack Detection for ROS 2 Robotic Systems: A Survey of Physics-Based Validation and Open Challenges* — DOI **10.1109/OJIES.2026.3731477** [141]

Bottom line

The most promising unexplored intersection is a **closed-loop, soft-bodied fish robot whose geometry and stiffness are optimized in simulation, whose control runs in a modern middleware stack, and whose success is judged by disturbance to the surrounding water and animals rather than only speed or tracking error** [140] [141] [142] [150]. The evidence suggests that the field is ready for a phased program from validated simulation to tank prototype to field trials, but not yet for claims of a fully integrated, ecologically validated system [64] [122] [139] [145].

Confidence level: High — multiple independent studies consistently support the gap between partial validation and full system integration, and they repeatedly identify the same bottlenecks in simulation fidelity, soft actuation, sensing, and field validation [139] [140] [141] [142].

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